Expressing algebraic numbers by radicals: a complete replacement for `ToRadicals`

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Abstract

Mathematica StackExchange question 34011 asks why `ToRadicals` cannot rewrite every `Root` object that has a radical expression, whether the obstruction is undecidability or cost, and whether a complete replacement can be written. We answer all three questions. A radical expression exists exactly when the Galois group of the minimal polynomial is solvable; this is decidable, and even in polynomial time, so the built-in limitation is a matter of scope and cost, not of principle. We give a complete algorithm: cheap structural recognizers that produce short formulas (functional decomposition, generalized reciprocal symmetry, Dickson polynomials, and the pair-sum resolvent that systematizes the search in the notebook attached to the question), and a general Galois–Kummer descent that is complete for every solvable input and proves nonsolvability otherwise. Every step is proved. Both examples of the question are worked out by hand and by machine, with a proof that the branch returned is the requested one. Two implementations are provided and tested: a Wolfram Language package and a Python port on python-flint, both reusing the splitting-field engine of the companion root-decomposition project. Every claim about running code in this article was executed; the measurements are reported as observed.

Contents

1 Introduction

The documentation of `ToRadicals` states that there are cases in which an expression in radicals exists but the function cannot find it. The question gives two such cases,

$$\alpha = \texttt{Root}[x^6 + x^4 - x^3 - x^2 - 1,\ 2], \qquad \beta = \texttt{Root}[5x^5 - 25x^3 + 25x + 6,\ 5], \tag{1}$$

with the closed forms

$$\begin{aligned} \alpha &= \frac{1}{\sqrt[3]{36}}\Big(\frac{3}{\sqrt{\gamma}} + \frac{\gamma}{2}\Big), \qquad \gamma = \sqrt[3]{2\delta} - 8\sqrt[3]{3/\delta}, \qquad \delta = 9 + \sqrt{849}, \\ \beta &= \sqrt[5]{\tfrac{-3+4i}{5}} + \sqrt[5]{\tfrac{-3-4i}{5}}, \end{aligned} \tag{2}$$

and asks: what is the nature of the restriction, is the problem undecidable or merely expensive, and can one write a version of `ToRadicals` that succeeds whenever a radical expression exists?

1.1 Summary of results

1. **Nature of the restriction.** An algebraic number has a radical expression if and only if the Galois group of its minimal polynomial is solvable (Theorem 4.1). Solvability of a Galois group is decidable, and Landau and Miller showed in 1985 that it is decidable in polynomial time [1]. The built-in function is documented to be complete through degree four and to recognize some special forms; on the two examples above and on the cyclic quintic $x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1$ (whose roots are $2\cos(2\pi k/11)$) it returns the `Root` object unchanged (Section 3). The obstruction is therefore neither undecidability nor the absence of an algorithm; it is that the general construction requires the splitting field, whose degree can be as large as the group order.
2. **A complete algorithm.** Section 7 gives the general Galois–Kummer descent with proofs and a completeness theorem (Theorem 7.3). Section 5 gives four structural reductions that produce short formulas without the splitting field, each with a proof; the two examples of the question are instances of two of them (Section 6). Section 8 gives cheap proofs of nonsolvability from Frobenius cycle types.

3. **Two implementations.** `RootToRadicals.wl` (Wolfram Language) and `roottoradicals.py` (Python, python-flint and SymPy) implement the same algorithms; both were executed on the examples and on regression suites (Section 11). Both examples of the question are solved in well under a second by the structural layer, and in 41 and 18 seconds (Wolfram) or 24 and 12 seconds (Python) when the general descent is forced.
4. **The notebook and the nine reports.** Section 9 explains what the notebook attached to the question computes and which theorem replaces each of its devices. Section 10 compares the nine independently written reports that preceded this article.

1.2 Conventions

A *radical expression* is built from rational numbers, the imaginary unit i, addition, multiplication, and powers with rational exponents, where $z^{p/q}$ denotes the principal branch $\exp(\frac{p}{q}\log z)$ with $-\pi < \operatorname{Im}\log z \le \pi$. This is exactly the grammar of Wolfram's `Plus`, `Times`, `Power` with rational exponents and `I`; in particular $\zeta_m = (-1)^{2/m} = e^{2\pi i/m}$ is a radical expression, so roots of unity need no special treatment. The *depth* of a radical expression is the nesting depth of its non-integer powers. Throughout, $p \in \mathbb{Z}[x]$ is the minimal polynomial of the input, $n = \deg p$, L is its splitting field in $\mathbb{C}$, and $G = \mathrm{Gal}(L/\mathbb{Q})$, viewed as a permutation group on the n roots. `Root`$[p, k]$ denotes the k-th root in Wolfram's ordering (real roots first, increasing; then non-real roots in conjugate pairs).

2 The problem

There are three distinct tasks.
1. *Decide* whether some radical expression equals the given number.
2. *Construct* one.
3. *Identify the requested conjugate*: the formula must equal the root with the given index, not merely some root of p.

The third task is not a formality. A formula in radicals with principal branches determines a single complex number, and a change of one branch usually gives a different conjugate; the notebook of the question spends most of its effort on exactly this point (Section 9). Our algorithms select branches by a certified numerical comparison among finitely many separated candidates and then verify the final identity exactly.

The criterion must be applied to the *minimal* polynomial of the selected number. If the user supplies a reducible or non-minimal annihilating polynomial, its Galois group may be larger (even nonsolvable) although the selected root lies in a solvable factor. All functions below therefore start by computing the minimal polynomial.

3 Why the built-in function fails, and why the problem is decidable

Wolfram's `ToRadicals` is documented to convert `Root` objects of degree at most four, and `Root` objects of some recognized special forms (for instance binomials $x^m - c$ and polynomials that `Decompose` reduces to such pieces). In Wolfram 15.0.1 we measured:

input	`ToRadicals`	remark
`Root`$[x^6+x^4-x^3-x^2-1,2]$	unchanged	example 1 of the question
`Root`$[5x^5-25x^3+25x+6,5]$	unchanged	example 2 of the question
`Root`$[x^5+x^4-4x^3-3x^2+3x+1,1]$	unchanged	cyclic quintic, $2\cos(2\pi/11)$
`Root`$[x^3-3x+1,1]$	radicals	degree ≤ 4
`Root`$[x^4-x-1,1]$	radicals	degree ≤ 4
`Root`$[x^5-2,1]$, `Root`$[x^8+1,1]$	radicals	binomials
`Root`$[x^6-2x^3-1,1]$	radicals	decomposable

None of the three failures is caused by undecidability. The Galois group of a polynomial is a finite computable object (by any of the classical resolvent methods, or by the numerical resolvent method of our engine, Section 11.1), and solvability of a finite group is decided by iterating the commutator subgroup. Landau and Miller [1] proved that solvability by radicals can be decided in time polynomial in the degree and the coefficient size, without computing the Galois group; the *construction* of an expression may however require the splitting field, whose degree $[L:\mathbb{Q}]=|G|$ can be $n!$. This is the real cost, and it is why a general-purpose library function stops at degree four: beyond it there is no bound on the size of the answer that is polynomial in the input.

4 The criterion

Theorem 4.1 (Galois). *Let $a\in\overline{\mathbb{Q}}$ with minimal polynomial p, splitting field L and $G=\mathrm{Gal}(L/\mathbb{Q})$. Then a is equal to a radical expression if and only if G is solvable.*

Proof. See [2, Chapter 5] or [3, §14.7]. For the direction we use constructively, Section 7 gives an explicit procedure. ☐

The constructive direction needs roots of unity in the base field. The following reduction is used by every algorithm in this article.

Lemma 4.2 (Cyclotomic base). *Let $m=\mathrm{rad}\,|G|$ be the product of the distinct primes dividing $|G|$, $B=\mathbb{Q}(\zeta_m)$, and $M=LB$ the compositum. Then*

(i) $H:=\mathrm{Gal}(M/B)\cong\mathrm{Gal}(L/L\cap B)$, a subgroup of G;

(ii) H is solvable if and only if G is solvable;

(iii) every prime dividing $|H|$ divides m, so B contains ζ_q for every composition factor q of H.

Proof. (i) is the theorem on natural irrationalities [2, Chapter 3]: restriction $\mathrm{Gal}(M/B)\to\mathrm{Gal}(L/L\cap B)$ is an isomorphism because $M=LB$ and $L/\mathbb{Q}$ is Galois. (ii) A subgroup of a solvable group is solvable; conversely $G/\mathrm{Gal}(L/L\cap B)\cong\mathrm{Gal}(L\cap B/\mathbb{Q})$ is a quotient of the abelian group $\mathrm{Gal}(B/\mathbb{Q})$, and an extension of a solvable group by an abelian group is solvable. (iii) follows from (i) and Lagrange's theorem. ☐

The lemma converts the problem into a tower of cyclic extensions of prime degree q over fields containing ζ_q, which is the setting of Kummer theory (Section 7). Note that m omits the prime 2 in practice: $\zeta_2=-1$ is rational.

5 Structural reductions

The reductions of this section find short formulas from the shape of p alone, without computing the splitting field. Each is a theorem of the form "if p has this shape, then a is obtained from a number of smaller degree by one explicit radical step". The implementations try them in the order given, recursing on the smaller number, and fall back to the general descent.

5.1 Functional decomposition

Proposition 5.1. *Let $p = g \circ h$ with $g, h \in \mathbb{Q}[x]$ of degrees ≥ 2, and let a be a root of p. Then $v = h(a)$ is a root of g, and a is a root of $h(x) - v \in \mathbb{Q}(v)[x]$. If v has a radical expression, then so does a whenever $\deg h \leq 4$ or $h(x) = cx^k + d$ is a binomial.*

Proof. $g(v) = g(h(a)) = p(a) = 0$. For $\deg h \leq 4$ the roots of $h(x) - v$ are given by the classical formulas with coefficients in $\mathbb{Q}(v)$; for a binomial they are $\zeta_k^e((v-d)/c)^{1/k}$. Both are radical expressions in a radical expression of v. $\square$

`Decompose` returns a complete decomposition $p = g_1 \circ g_2 \circ \cdots \circ g_r$ (Ritt's theorem guarantees that the multiset of degrees is well defined). The implementation peels from the outside: v_1 is a root of g_1, then v_2 is a root of $g_2(x) - v_1$, and so on; at each stage the exact value $v_i = (g_{i+1} \circ \cdots \circ g_r)(a)$ is known (it is an algebraic number computed by `RootReduce` or by a resultant), and the branch of the radical formula is the candidate that matches this exact value numerically. The chain fails only when some inner piece of degree > 4 is not a binomial; the general descent then takes over.

5.2 Generalized reciprocal symmetry

Proposition 5.2. *Let $n = 2m$ and suppose $p(x) = x^m P(x + c/x)$ with $c \in \mathbb{Q}^\times$ and $P \in \mathbb{Q}[x]$ of degree m. If a is a root of p, then $y = a + c/a$ is a root of P, and $a = \frac{1}{2}(y \pm \sqrt{y^2 - 4c})$; the other sign gives the conjugate c/a.*

Proof. The constant term of the monic form of p is $c^m \neq 0$, so $a \neq 0$ and $0 = p(a) = a^m P(a + c/a)$ gives $P(y) = 0$. The numbers a and c/a are the two roots of $x^2 - yx + c$. $\square$

Detection. Write p monic. Then c^m is the constant term, so c is a rational m-th root of it (two candidates $\pm c$ when m is even); for each candidate, subtract $\mathrm{coef}_{m+k}(p)\, x^m(x + c/x)^k$ for $k = m, m-1, \ldots, 0$ and check that the remainder vanishes. This is a finite exact test.

Branch coupling. The formula carries c/a as the other root of the same quadratic. Extracting a and c/a by two independent radical formulas and then choosing branches for each would not be sound: the two choices are coupled by $a \cdot (c/a) = c$. The same remark applies to Dickson polynomials below, where the term c/u must be formed from the *same* u. The first example of the question (Section 6) is of this type with $c = -1$.

5.3 Dickson polynomials

The Dickson polynomial $D_k(x, c)$ is defined by $D_0 = 2$, $D_1 = x$, $D_k = xD_{k-1} - cD_{k-2}$, and satisfies the functional equation

$$D_k\Big(u + \frac{c}{u},\, c\Big) = u^k + \frac{c^k}{u^k}. \tag{3}$$

Proposition 5.3. *Suppose that after the translation $x \mapsto x + t$ that kills the x^{n-1} term, the monic form of p becomes $D_n(x, c) - b$ with $c \in \mathbb{Q}^\times$, $b \in \mathbb{Q}$. Let s be a root of $s^2 - bs + c^n = 0$ and $u = s^{1/n}$. Then the n numbers*

$$t + \zeta_n^j u + \frac{c}{\zeta_n^j u}, \qquad j = 0, \ldots, n-1,$$

are roots of p; if they are distinct they are all of them.

Proof. With $s' = c^n/s$ we have $s + s' = b$. By (3), $D_n(\zeta^j u + c/(\zeta^j u), c) = (\zeta^j u)^n + c^n/(\zeta^j u)^n = s + s' = b$. $\square$

Detection. $D_n(x, c) = x^n - ncx^{n-2} + \ldots$, so c is read off the x^{n-2} coefficient of the translated polynomial and the identity $p = D_n(x, c) - b$ is checked exactly. The second example of the question is $5(D_5(x, 1) + \frac{6}{5})$.

5.4 The pair-sum resolvent

The following identity is classical.

Lemma 5.4. *Let p have leading coefficient ℓ and roots $a_1, \dots, a_n$, and let $R_2(y) = \prod_{i<j}(y - a_i - a_j)$. Then*

$$\mathrm{Res}_x(p(x),\, p(y-x)) = 2^n \ell^{2n-1}\, p(y/2)\, R_2(y)^2.$$

Proof. $\mathrm{Res}_x(p(x), p(y-x)) = \ell^n \prod_i p(y-a_i) = \ell^{2n} \prod_{i,j}(y - a_i - a_j)$. The diagonal terms give $\prod_i (y - 2a_i) = 2^n \ell^{-1} p(y/2)$; the off-diagonal terms give $R_2(y)^2$. □

So R_2 is computed exactly by a resultant, an exact division, and a square root of a factorization. Its roots are the pair sums $a_i + a_j$.

Proposition 5.5. *Let $y_0 = a + a'$ be a pair sum with a' a conjugate of a, and let $K = \mathbb{Q}(y_0)$. Factor p over K and let $f \in K[x]$ be the irreducible factor with $f(a) = 0$. If y_0 has a radical expression and $\deg f \le 4$ (or f is a binomial over K), then a has a radical expression, obtained by substituting the expression of y_0 into the coefficients of f and solving.*

Proof. Immediate from the classical formulas, as in Proposition 5.1. □

In practice one takes the factors of R_2 over $\mathbb{Q}$ of degree $2 \le m < n$, in increasing degree, tests numerically which of their roots are of the form $a + a'$, factors p over $\mathbb{Q}(y_0)$ (Wolfram's `Factor` with `Extension` is fast for the small degrees involved) and recurses on y_0. When $a' = c/a$ this reproduces Proposition 5.2, but the pair-sum reduction is more general: the product aa' need not be rational, it is a polynomial in y_0 read off the factor f. Section 9 shows that this is exactly what the notebook's `findExtension` searches for by elimination.

6 The two examples of the question

6.1 The sextic

$p(x) = x^6 + x^4 - x^3 - x^2 - 1$ is monic with constant term -1, so in Proposition 5.2 $m = 3$ and $c^3 = -1$, $c = -1$. Peeling gives

$$p(x) = x^3\Big(\big(x - \tfrac{1}{x}\big)^3 + 4\big(x - \tfrac{1}{x}\big) - 1\Big), \qquad P(y) = y^3 + 4y - 1.$$

Cardano's formula for the real root of P (discriminant $-4 \cdot 4^3 - 27 < 0$, one real root) gives

$$y = u - \frac{4}{3u}, \qquad u = \Big(\frac{9 + \sqrt{849}}{18}\Big)^{1/3},$$

and $\alpha = \frac{1}{2}(y + \sqrt{y^2+4})$. This is the expression the package returns (in Wolfram's own arrangement of the cube roots), and it is equal to the closed form (2) of the question: with γ as there, $\gamma/\sqrt[3]{36} = y$ and $3/(\sqrt[3]{36}\sqrt{\gamma}) = \frac{1}{2}\sqrt{y^2+4} - \frac{y}{2}$, as one checks by squaring; `RootReduce` confirms the identity.

Branch. p has exactly two real roots, $\alpha_1 < 0 < \alpha_2$, and $\texttt{Root}[p, 2] = \alpha_2$ is the positive one. The map $x \mapsto x - 1/x$ is increasing on $(0, \infty)$ and on $(-\infty, 0)$, and y is the unique real root of P; the two real solutions of $x - 1/x = y$ are $\frac{1}{2}(y \pm \sqrt{y^2+4})$, of which the one with the plus sign is positive. Hence the formula equals $\texttt{Root}[p, 2]$; the minus sign gives $\texttt{Root}[p, 1] = -1/\alpha_2$. The four non-real roots come from the two non-real roots of P, which are the other two branches of the cube root in u.

6.2 The quintic

$p(x) = 5x^5 - 25x^3 + 25x + 6 = 5(x^5 - 5x^3 + 5x + \frac{6}{5})$, and $D_5(x,1) = x^5 - 5x^3 + 5x$, so $p = 5\,(D_5(x,1) - b)$ with $b = -6/5$ and $c = 1$. The quadratic $s^2 + \frac{6}{5}s + 1 = 0$ has roots $s = \frac{-3\pm 4i}{5}$, of modulus 1. By Proposition 5.3 the roots are $\zeta_5^j u + \zeta_5^{-j} u^{-1}$ with $u = s^{1/5}$; since $|s| = 1$, $u^{-1} = \bar{u}$ and the roots are $2\,\mathrm{Re}(\zeta_5^j u) = 2\cos(\frac{\theta+2\pi j}{5})$, $\theta = \arg s$, all real. The package returns $s^{1/5} + s^{-1/5}$ with $s = \frac{-3+4i}{5}$, which is (2) because $s^{-1/5} = \bar{s}^{1/5}$ for $|s| = 1$ (both are the principal fifth root of $\bar{s}$: the principal argument of $\bar{s}$ is $-\theta$ and $s^{-1/5} = e^{-i\theta/5}$).

Branch. $\theta = \arg \frac{-3+4i}{5} \in (\pi/2, \pi)$, so $\theta/5 \in (\pi/10, \pi/5)$ and $2\cos(\theta/5)$ is the largest of the five values $2\cos((\theta + 2\pi j)/5)$; all five roots are real, so `Root`$[p, 5]$ is the largest root, which is the returned branch $j = 0$. The other conjugates are $j = 1, \dots, 4$, and the package returns each of them for the corresponding index.

7 The general Galois–Kummer descent

7.1 Setting

Let $B = \mathbb{Q}(\zeta_m)$ and $M = LB$ as in Lemma 4.2, and $H = \mathrm{Gal}(M/B)$. Choose a composition series

$$H = H_0 \triangleright H_1 \triangleright \cdots \triangleright H_r = 1, \qquad |H_{i-1}/H_i| = q_i \text{ prime},$$

which exists exactly when H is solvable (the implementation builds it from the commutator subgroup: H_i is a subgroup of prime index in H_{i-1} containing $[H_{i-1}, H_{i-1}]$, hence normal). Let $F_i = M^{H_i}$ be the fixed fields, so $B = F_0 \subset F_1 \subset \cdots \subset F_r = M$ with F_i/F_{i-1} cyclic of prime degree q_i, generated by the image of any $\sigma_i \in H_{i-1} \setminus H_i$.

7.2 One Kummer step

Proposition 7.1 (Lagrange resolvents)**.** *Let $F \subset E$ be fields with E/F cyclic of prime degree q, generated by σ, and suppose $\zeta = \zeta_q \in F$. For $v \in E$ put*

$$R_k = \sum_{j=0}^{q-1} \zeta^{-kj} \sigma^j(v), \qquad k = 0, \dots, q-1.$$

Then

(i) $\sigma(R_k) = \zeta^k R_k$; hence $R_k^q \in F$, and $R_0 \in F$;

(ii) $\sum_{k=0}^{q-1} R_k = q\,v$;

(iii) if $R_{k_1} \neq 0$ for some $k_1 \neq 0$, then for every k the element $c_k = R_k/R_{k_1}^m$ with $m \equiv k\,k_1^{-1}$ (mod q) lies in F;

(iv) if $v \notin F$ then some R_k with $k \neq 0$ is nonzero.

Proof. (i) $\sigma(R_k) = \sum_j \zeta^{-kj}\sigma^{j+1}(v) = \zeta^k \sum_{j'} \zeta^{-kj'}\sigma^{j'}(v)$, using $\sigma^q = 1$ and $\zeta^q = 1$. Then $\sigma(R_k^q) = \zeta^{kq} R_k^q = R_k^q$, so R_k^q is fixed by $\mathrm{Gal}(E/F)$. (ii) $\sum_k R_k = \sum_j (\sum_k \zeta^{-kj})\sigma^j(v)$ and the inner sum is q for $j = 0$ and 0 otherwise. (iii) $\sigma(c_k) = \zeta^k R_k/(\zeta^{k_1 m} R_{k_1}^m) = c_k$ since $k_1 m \equiv k$. (iv) If all R_k with $k \neq 0$ vanish, (ii) gives $qv = R_0 \in F$. ☐

Applied to $F = F_{i-1}$, $E = F_i$, $\sigma = \sigma_i$ and an element $v \in F_i$, the proposition expresses v through elements of F_{i-1} and q-th roots of elements of F_{i-1} in one of two ways:

$$\text{(Fourier form)} \qquad v = \frac{1}{q}\Big(R_0 + \sum_{k\neq 0} \zeta^{e_k}\,(R_k^q)^{1/q}\Big), \tag{4}$$

$$\text{(eigenvector form)} \qquad v = \frac{1}{q}\Big(R_0 + u + \sum_{k \neq 0, k_1} c_k\, u^{m_k}\Big), \quad u = \zeta^e (R_{k_1}^q)^{1/q}. \tag{5}$$

The exponents e_k, e are the *branches*: $(R_k^q)^{1/q}$ is the principal root, and exactly one of the q numbers $\zeta^e(R_k^q)^{1/q}$ equals R_k. The Fourier form extracts $q-1$ radicals at this level; the eigenvector form extracts one and pushes the ratios c_k, which are elements of F_{i-1}, one level down. Which form is shorter depends on the case (for the cyclic quintic the Fourier form has 140 leaves in Wolfram's count, the eigenvector form 173; for cubics both forms coincide with Cardano's). The implementations compute both and keep the shorter, or use the one requested.

7.3 Branch selection

The candidates $\zeta^e Q^{1/q}$, $e = 0, \ldots, q-1$, for $Q = R_k^q \neq 0$ are the q distinct q-th roots of Q; they are separated by at least $|Q|^{1/q} \cdot 2\sin(\pi/q)$. The value of R_k is known independently from the exact coordinates of R_k in the splitting field (Section 11.1), so a numerical comparison at sufficient precision identifies e. In Wolfram the comparison is made at 60 digits with a separation check, and the final identity is verified exactly by `RootReduce`; in Python it is made in ball arithmetic and is rigorous by itself: a candidate is accepted only if it is the unique one whose ball overlaps the ball of R_k (the true candidate always overlaps, so uniqueness identifies it), and the precision is doubled until this happens.

The branch cut. There is one trap in rigorous evaluation that none of the nine reports mentions. When Q is a real negative number that is *written* as a sum of complex-conjugate radicals (which happens as soon as a real number is expressed through non-real radicands, for instance every real root of a cubic with three real roots), its computed enclosure is a small box straddling the negative real axis. The principal q-th root is discontinuous there, so the enclosure of $Q^{1/q}$ covers both limits and no precision separates the candidates. The exact data resolve this: Q is real if and only if it is fixed by the automorphism that acts as complex conjugation on the roots (a permutation the engine identifies), and its sign is then certified from the real part. In that case the implementation writes the principal root as $(-1)^{1/q}(-Q)^{1/q}$, an identity of principal branches for real negative Q, and the enclosure of $-Q$ lies near the positive axis where the root is continuous. The same rewriting is applied in the structural layer to discriminants and to binomial right-hand sides, whose exact values are known as algebraic numbers.

7.4 The algorithm and its completeness

Algorithm 7.2. Input: $a = \mathtt{Root}[p, k]$ with p irreducible.

1. Compute G as a permutation group on the roots, with the splitting field L in coordinates (Section 11.1). If G is not solvable, stop: no radical expression exists.
2. Let m be the product of the odd primes dividing $|G|$. Compute the same data for $p \cdot \prod_{q|m} \Phi_q$ (omitting a factor equal to p); this is $M = L(\zeta_m)$ with its group. Locate ζ_q among the roots and let H be the stabilizer of all of them.
3. Build a composition series of H with prime quotients, a generator σ_i of each quotient, and the coordinates of a.
4. Define $\rho(v, i)$ for $v \in F_i$ recursively: if $i = 0$, solve for the coordinates of v in the basis $\{\prod_q \zeta_q^{e_q} : 0 \leq e_q \leq q-2\}$ of B and return the corresponding expression in $(-1)^{2/q}$; if v is fixed by H_{i-1}, return $\rho(v, i-1)$; otherwise form the R_k of Proposition 7.1, compute R_k^q and the c_k by exact linear algebra in coordinates, select branches as in Section 7.3, and return (4) or (5) with $\rho(\cdot, i-1)$ applied to R_0, R_k^q and c_k. Memoize on (v, i).
5. Output $\rho(a, r)$ and verify $\rho(a, r) - a = 0$ exactly.

Theorem 7.3 (Completeness)**.** *Algorithm 7.2 terminates. If G is solvable it outputs a radical expression equal to* $\mathtt{Root}[p, k]$ *of depth at most $r+1$, where r is the length of the composition series of H; otherwise it reports that no radical expression exists.*

Proof. Nonsolvability is decided exactly by the commutator series. If G is solvable, H is solvable by Lemma 4.2(ii) and the series exists; by Lemma 4.2(iii), $\zeta_{q_i} \in B \subseteq F_{i-1}$ for every i, so

Proposition 7.1 applies at every level. The recursion $\rho(v, i)$ calls ρ only with second argument $i-1$ and terminates. Correctness: at level 0 the element lies in B and its expression in the power basis of the ζ_q is exact; at level i, the identity (4) or (5) holds exactly in M with the true branches, and the true branches are the selected ones because the candidates are separated and the comparison is made at a precision exceeding the separation (rigorously so in ball arithmetic). Each level adds at most one nesting of radicals to the expressions coming from the level below, and the base expressions have depth one, so the depth is at most $r+1$. Finally, the exact verification in step 5 is an independent check. □

Remark 7.4 (Cost). The cost is dominated by step 2: the splitting field of $p \prod \Phi_q$ has degree $|H| \cdot \varphi(m)$ over $\mathbb{Q}$, and our engine builds it by adjoining one root at a time with `RootReduce`; for the two examples of the question the extended groups have orders 48 and 40, computed in about 33 and 10 seconds in Wolfram. The group order limit (`"MaxGroupOrder"`, default 400) turns larger cases into an explicit `ResourceLimit` status rather than a wrong answer. The only heuristic component is the choice of random weights in the resolvent construction, which is checked by exact counting and closure tests.

7.5 Two refinements

The tensor algebra. Several reports adjoin ζ_q by working in $L[z]/\Phi_q(z) \cong L \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_q)$ instead of in the field $L(\zeta_q)$. This is a ring, not a field, as soon as $\zeta_q \in L$ (Φ_q splits in L), and it has zero divisors; report 05 observes this and makes its descent division-free. Our implementations avoid the issue by computing the splitting field of $p \prod \Phi_q$ directly, which is a field whose degree is the true $[M : \mathbb{Q}]$ (for $x^3 - 2$, adjoining Φ_3 does not enlarge the field, and the engine sees a group of order 6, not 12). The one division we perform, $c_k = R_k / R_{k_1}^m$, is in a field and is exact.

Fourier versus eigenvector. The Fourier form (4) needs no nonzero-generator search (report 03), the eigenvector form (5) usually gives shorter formulas (report 04). The two forms are equally valid and we implement both; the eigenvector form needs the extra exactness argument of Proposition 7.1(iii), which we have supplied.

8 Negative results without the group

Computing G exactly is the expensive step (for $x^5 - x - 1$, whose group is S_5 of order 120, our engine needs 18 minutes in Wolfram). Two classical facts give nonsolvability proofs in milliseconds.

Theorem 8.1 (Dedekind–Frobenius)**.** *Let ℓ be a prime not dividing the discriminant or the leading coefficient of p. If $p \bmod \ell$ factors into irreducible factors of degrees $d_1, \ldots, d_s$, then G contains a permutation of cycle type $(d_1, \ldots, d_s)$.*

Theorem 8.2 (Galois)**.** *A solvable transitive permutation group of prime degree n is conjugate to a subgroup of $\mathrm{AGL}(1, n) = \{x \mapsto ax + b\}$ acting on $\mathbb{F}_n$. Consequently each of its elements has cycle type 1^n, n, or $1\, d^{(n-1)/d}$ for some divisor d of $n-1$.*

Proof. The first statement is Galois' theorem on solvable transitive groups of prime degree [2, Chapter 4], [3, §14.7]. For the second, $x \mapsto x + b$ with $b \neq 0$ is an n-cycle; $x \mapsto ax + b$ with $a \neq 1$ has the unique fixed point $x = b/(1-a)$ and, after conjugating by a translation, is $x \mapsto ax$, whose nonzero orbits all have length $\mathrm{ord}(a) \mid n - 1$. □

Corollary 8.3. *If n is prime and some Frobenius cycle type is not of the form in Theorem 8.2, then G is not solvable and* `Root`*$[p, k]$ has no radical expression.*

For example $x^5 - x - 1 \bmod 3$ has factors of degrees $(2, 3)$ and $x^7 - x - 1 \bmod 3$ has degrees $(1, 2, 4)$; neither is admissible, so both are settled immediately.

For composite degrees we use a second criterion.

Proposition 8.4. *Let n be composite and not a prime power. If G contains a cycle of prime length $\ell > n/2$ (with all other points fixed), then G is not solvable.*

Proof. A transitive group containing an ℓ-cycle with $\ell > n/2$ is primitive: if Δ were a block system with blocks of size $1 < b < n$, the ℓ-cycle would permute the blocks it meets in a cycle of length dividing ℓ; length 1 would force its support of size ℓ into a single block of size $b \leq n/2 < \ell$, and length ℓ would need $\ell b \leq n$, impossible for $b \geq 2$. A solvable primitive group has a unique minimal normal subgroup, which is elementary abelian and regular, so its degree is a prime power [4, Chapter II]. Since n is not a prime power, G is not solvable. □

Both criteria are inconclusive in the other direction (they never prove solvability), and neither applies to composite prime-power degrees $4, 8, 9, 16, \ldots$; in those cases the implementations compute the group exactly, subject to the group order limit. For $x^6 - x - 1$ (degree 6, group S_6) and $x^6 + x^4 + 3x^2 - 2x + 5$ (group of order 120) a 5-cycle is found within the first few primes and the Frobenius test proves nonsolvability in about 10 milliseconds (a full `RootToRadicals` call takes up to a second more, spent in the built-in `ToRadicals` attempt that precedes it).

9 What the notebook of the question does

The notebook `FindExtension.nb` attached to the question (an inert transcription of its input cells is kept in `reports/report-09/original/`) is not about the two examples above. Its target is

$$\beta_0 = I^{-1}_{1/9}(\tfrac{1}{3}, \tfrac{1}{3}),$$

the inverse of the regularized incomplete beta function, which the author evaluates to 1000 digits and recognizes with `RootApproximant` as an algebraic number; the notebook then tries to express that number through nested radicals and $\cos(\pi/9)$. We reproduced the recognition: `RootApproximant` returns in 7 seconds a monic integer polynomial of degree 27 (leading terms $x^{27} + 108x^{26} - 2970x^{25} + \ldots$, constant term 1), the identity holds to 400 digits, and over $\mathbb{Q}(3^{1/6})$ the polynomial factors into pieces of degrees 9 and 18, which is the tower the notebook finds. Since $27 = 3^3$ is a prime power, neither Frobenius test of Section 8 applies, and the exact group of the degree-27 polynomial did not finish within 15 minutes in Wolfram even with the order limit raised to 3000. The notebook's tower settles the question nevertheless: the degree-9 number θ that the notebook adjoins (the third root of $x^9 - 657x^8 + 6111x^7 + 3318x^6 + 19647x^5 - 12033x^4 + 3972x^3 - 684x^2 + 9x - 1$) has a Galois group of order 18, and `RootToRadicals` expresses it in radicals by two Kummer steps of prime 3 in 43 seconds (verified exactly; depth 3, 111 leaves); the degree-9 factor of the minimal polynomial over $\mathbb{Q}(3^{1/6})$ that vanishes at β_0 splits over $\mathbb{Q}(3^{1/6}, \theta)$ into three cubics (237 seconds), so β_0 is a root of a cubic with radical coefficients and therefore has a radical expression. We have not assembled that expression. The notebook's devices, in the order they appear, are:

`radicalDepth` the nesting depth of rational powers (and of `Sin`, `Cos`), the same measure as our `RadicalDepth`.

`findExtension` eliminates p, q from `PolynomialRemainder`$[mp,\ x^2 + px + q] = 0$ and returns the simplest p with a solution. A monic quadratic dividing the minimal polynomial has $-p = a_i + a_j$, so this is a search over the pair sums, i.e. over the roots of the pair-sum resolvent R_2 of Lemma 5.4, with ``Simplify`SimplifyCount`` as the selection criterion. `findExtension3` does the same with cubic factors $x^3 + px^2 + qx + r$, i.e. with triple sums. The systematic replacement is Proposition 5.5: compute R_2 exactly, factor it over $\mathbb{Q}$, and factor p over $\mathbb{Q}(y_0)$ for a low-degree root y_0.

`project` tests whether the real or imaginary part of some conjugate has smaller degree than the number itself. This is a special case of the same idea: $2\,\mathrm{Re}(a) = a + \bar{a}$ is a pair sum.

`factor` factors the minimal polynomial over an extension and picks the factor vanishing at the number with `PossibleZeroQ`; this is the factorization step of Proposition 5.5.

solve picks, among the `Solve` roots of a low-degree factor, the one equal to the number (`PossibleZeroQ` again): branch selection (Section 7.3) done numerically.

fuzz draws random subsets of the conjugates, sums them, guesses a low-degree algebraic number with `RootApproximant` and checks it with `PossibleZeroQ`. This is a randomized search for elements of proper subfields of the splitting field; the Galois-theoretic replacement is the list of subgroups and fixed fields that the engine computes, or, for the specific purpose of radicals, the composition series of Section 7, which needs no search at all.

simplify iterates `FullSimplify` with a custom complexity function that penalizes `ArcTan`, hyperbolic functions and every `Root` object except one chosen degree-9 number. This is the only device without a systematic replacement: the shortness of a radical expression is not a Galois-theoretic invariant, and the reports that discuss it (04, 09) offer heuristics only.

The notebook's approach is sound as a discovery procedure and it does find the tower $\mathbb{Q} \subset \mathbb{Q}(3^{1/6}) \subset \mathbb{Q}(3^{1/6}, \theta)$ over which the target becomes a root of a cubic; what it lacks is a termination guarantee and a criterion for when to stop looking. The descent of Section 7 supplies both, at the price of the splitting field, and for the piece θ that the notebook could not simplify it produces the radical expression directly.

10 Comparison of the nine reports

Nine reports on this question were written independently before this article; they are kept verbatim in `reports/report-01` to `report-09`. All nine share the same core: the criterion on the minimal polynomial, the cyclotomic base with $m = \operatorname{rad}|G|$, a prime composition series, Lagrange or Fourier resolvents, exact branch selection, the same two closed forms for the examples, and a completeness theorem conditional on exact primitives. Table 1 lists what each one adds.

All nine reports state the same two closed forms for the examples (report 08 writes the sextic with two positive cube roots, $((9+\sqrt{849})/18)^{1/3} - ((\sqrt{849}-9)/18)^{1/3}$, which is the same number). Three reports (02, 03, 07) ship a native Wolfram implementation of the general algorithm; none of the three was executed by its author. The only report that forced the general descent on the original examples (01) did not finish, because it built the splitting field in the power basis of a primitive element (Section 11.1). Four reports (01, 04, 05, 09) executed a prime-5 Kummer step on some input. The status taxonomy of report 09 is adopted here, with `SearchExhausted` renamed `NotFound`.

Three points needed correction or independent verification. Report 07 cites Milne "Theorem 3.28" for the solvability criterion; the theorem is in Chapter 5 of [2]. Several reports refer to `MANIFEST` or `SHA256SUMS` files that are not present here (checksum ledgers are not kept in this repository). And no report executed any Wolfram Language code: every Wolfram claim in them was static. We ran the Wolfram side natively; the results are in Section 11.3.

Every report proposes to build the splitting field with `ToNumberField[roots, All]` (or its SymPy analogue `primitive_element`). The companion project measured that call at 14 minutes for a degree-36 field, and the reason is structural: the power basis of a generic primitive element has coefficients of enormous height. Our implementations use the tower basis with trace-form coordinates of the companion engine instead (Section 11.1), which is why the forced descent on the examples takes seconds rather than failing to finish, as it did in report 01.

11 Algorithms and implementation

11.1 The splitting-field engine

Both implementations reuse the Galois engine of the companion project (`RootDecomposition.wl` and `rootdecomp.py` in `root-decomposition/`). It computes the Galois group by numerical resolvents: roots are adjoined one at a time, $\theta_k = \theta_{k-1} + w_k a_k$ with small random integer

report	distinctive contribution	what was executed
01	Kummer generators as a rational nullspace; "rational rectangle" embeddings with exact sign certificates.	The only forced general run on the two original examples; neither finished within its 480-second limit (the sextic reached the degree-48 field, the quintic the degree-40 field). Two prime-5 Kummer steps (cyclic and binomial quintic, 22 and 28 s).
02	The pair-sum resolvent $\mathrm{Res}_x(f(x), f(y-x)) = 2^n f(y/2) R_2(y)^2$ identified as the systematic form of the notebook's `findExtension`; a trace–Fourier basis search with a termination proof; a native Wolfram `kummerSolve`.	Structural paths on the examples (0.7–2.6 s per root); forced descent only on $x^3 - 2$, $x^4 - 2$.
03	Fourier inversion $a = \frac{1}{p} \sum_j R_j$ needing no generator search; the clearest account of SymPy embedding pitfalls; the point that an oversized supplied field proves nothing about nonsolvability; a native Wolfram `galoisConvert`.	Structural paths on the examples (1.8 s, 1.2 s).
04	Relative-automorphism recovery without a second factorization; trace-projector scoring for small radicands; an independent arithmetic-tower checker.	Structural paths on the examples (0.7 s, 0.2 s); the cyclic quintic by descent (16.7 s).
05	Division-free tensor algebra $L \otimes \mathbb{Q}(\zeta_p)$ with zero divisors (with a test multiplying two of them); resultant primitive elements with a square-free certificate; norm-polynomial justification of branch comparisons.	Structural paths (0.6 s, 0.2 s); the cyclic quintic by Fourier descent (1.0 s); a forced S_4 run hit its resource limit at 57 s.
06	Exact Frobenius negative test; a principal-sector branch predicate with conjugation-based zero tests; straight-line register programs as certificates.	Structural paths on all conjugates of both examples (up to 104 s for one sextic root); the forced cyclic quintic timed out at 90 s.
07	A native Wolfram `SystematicRadicals.wl` with Kummer eigenprojections; a notebook audit with line numbers.	Structural paths (2.3 s, 1.6 s); its forced run on $x^5 - 2$ was inconclusive at 30 s.
08	The coherent-branch theorem (any consistent branch assignment reproduces the whole root set); a purely rational certificate checker that proves the root set rather than an index.	Structural paths (milliseconds for the selected roots).
09	A five-status taxonomy (`Success`, `NotSolvable`, `SearchExhausted`, `ResourceLimit`, `VerificationFailure`); replayable certificates with an embedding anchor; a line-numbered notebook critique.	Structural paths (4.0 s, 0.3 s); forced descents on $x^5 - 2$ (34 s) and a nonabelian relative group (39 s).

Table 1: What each of the nine reports adds to the common core, and what its Python code actually executed (its own reported timings).

weights, the exact minimal polynomial of θ_k is found with RootReduce (Wolfram) or by rigorous rounding of a product of balls (Python), and the orbit of partial tuples is matched to its roots numerically with counting checks; the result is verified by group closure. The splitting field is represented in the tower monomial basis $\prod_j a_{g_j}^{e_j}$, and the coordinates of an element are obtained from its numerical conjugates through the trace form (the Gram matrix of the basis is an integer matrix), so multiplication matrices, automorphism matrices and fixed fields are exact rational data. Elements of M are therefore rational vectors, R_k, R_k^q and c_k are computed by exact matrix arithmetic, and the numerical value of any element is available at any precision.

11.2 Interface

```
Get["RootToRadicals.wl"];
RootToRadicals[Root[-1 - #^2 - #^3 + #^4 + #^6 &, 2]]
RootRadicalReport[Root[6 + 25 # - 25 #^3 + 5 #^5 &, 5]]
RootRadicalReport[Root[#^5 + #^4 - 4 #^3 - 3 #^2 + 3 # + 1 &, 1], Method -> "Galois"]
RootSolvableQ[Root[#^5 - # - 1 &, 1]]         (* False, by Frobenius cycle types *)
```

`RootToRadicals[a]` returns a radical expression or a `Failure` whose tag is one of `NotSolvable` (proved), `NotFound` (structural search exhausted, only with `Method -> "Structural"`), `ResourceLimit`, `VerificationFailed`, `Inexact`, `NotAlgebraic`, `InvalidOptions`. `RootRadicalReport` returns an association with the expression, the method (`ToRadicals`, `Decompose`, `Reciprocal`, `Dickson`, `PairSum`, `Galois`), the verification status (`True` by `RootReduce`, or `Indeterminate` when the exact check exceeded its time limit and only a 200-digit numerical check passed), the depth, the leaf count, the group orders and the primes of the composition series, and the timing. Options: `Method` (`Automatic`, `"Structural"`, `"Galois"`), `"Resolvents"` (`Automatic`, `"Fourier"`, `"Eigenvector"`), `"MaxGroupOrder"`, `"WorkingPrecision"`, `"VerificationTimeLimit"`, `"MaxDepth"`. `RadicalExpressionQ` and `RadicalDepth` implement the grammar of Section 1.2.

```
import roottoradicals as rt
r = rt.root_to_radicals("Root[-1 - #^2 - #^3 + #^4 + #^6 &, 2]")
r.expression, r.method, r.verified, r.wolfram()
rt.root_to_radicals("Root[1 + 3 # - 3 #^2 - 4 #^3 + #^4 + #^5 &, 1]", method="galois
    ")
rt.is_solvable("Root[-1 - # + #^5 &, 1]")      # False
```

The Python function returns a `RadicalResult` (a SymPy expression with the same metadata) or raises `NotSolvable`, `NotFound`, `ResourceLimit` or `PrecisionError`. Its structural layer has the classical formulas for degree ≤ 4, `Decompose`, reciprocal symmetry and Dickson polynomials; the pair-sum reduction, which needs factorization over a number field, is Wolfram-only, and the general descent covers those cases in Python. `verified` is a rigorous ball check that the expression encloses the requested root and no other conjugate; `python/verify_wolfram.py` additionally verifies Python's expressions exactly in a Wolfram kernel with `RootReduce`.

11.3 Measurements

All timings were measured on one Windows machine, in Wolfram 15.0.1 (`wolfram -script`) and in Python 3.14 with python-flint 0.8 and SymPy 1.14, with the default options (`"MaxGroupOrder"` 400, 80 digits or 300 bits). "Orders" are $|G|$ and the order of the group of $p \prod \Phi_q$.

input	method	orders	Wolfram	Python
sextic example α	Reciprocal		0.12 s	0.07 s
quintic example β	Dickson		0.04 s	0.01 s
sextic example, descent forced	Galois	$24 \to 48$	41 s	24 s
quintic example, descent forced	Galois	$20 \to 40$	18 s	12 s
cyclic quintic $x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1$	Galois	$5 \to 20$	8.4 s	0.37 s
the same, Fourier form only	Galois	$5 \to 20$	4.8 s	0.05 s
$x^4 - x - 1$, descent forced	Galois	$24 \to 48$	32 s	12 s
$x^5 - 2$, descent forced	Galois	$20 \to 20$	4.8 s	1.6 s
Φ_7, descent forced	Galois	$6 \to 12$	1.5 s	0.24 s
cyclic quintic composed with x^2 (degree 10)	Decompose	$5 \to 20$	9.3 s	0.19 s
$D_7(x, 1) - 3$	Dickson		0.10 s	0.02 s
sextic with pair sums of degree 3	PairSum / Galois*	$12 \to 24$	0.19 s	0.83 s
$x^5 - x - 1$: not solvable	Frobenius		1.2 s	0.26 s
$x^6 - x - 1$: not solvable	Frobenius		0.01 s	0.01 s
$x^5 - x - 1$ by the exact group (order 120)	group	120	1088 s	—

* The pair-sum reduction is Wolfram-only; Python solves this input by the descent.

The Wolfram times for the descent are dominated by the construction of the extended splitting field (about 33 s of the 41 s for the sextic); Python's Arb-based engine is faster there. The Wolfram time for the cyclic quintic with the default `"Resolvents" -> Automatic` includes computing both forms; the eigenvector form is longer for this input (173 versus 140 leaves) and is discarded. Wolfram's structural layer uses the built-in `ToRadicals` first (which handles degree ≤ 4, binomials and decomposable inputs in milliseconds), so its `Decompose` path is only reached for compositions whose outer piece the built-in function cannot solve, as in the degree-10 example.

The regression suites were executed: `RunTests.wl` runs 60 tests in 90 s of wall time (including the forced descents on both examples and all their conjugates); `test_roottoradicals.py` runs 46 cases in 51 s. `verify_wolfram.py` re-verifies 13 Python expressions exactly in a Wolfram kernel with `RootReduce`; all pass.

11.4 Limits and scope of the certificates

- A returned expression is exactly equal to the input: in Wolfram by `RootReduce`; in Python by the exactness of the coordinate identities together with rigorous branch selection, and again by the final ball check. The status `Verified -> Indeterminate` (Wolfram) is reported honestly when the exact check hits its time limit.
- `NotSolvable` is a proof: either a Frobenius cycle type (Corollary 8.3, Proposition 8.4) or the exact commutator series of the computed group. The group computation relies on numerical matching with counting and closure checks; its exactness has the same standing as in the companion project.
- `ResourceLimit` asserts nothing about solvability. Nonsolvable polynomials of composite prime-power degree with a group larger than the limit end here; so do solvable ones with $|H|\varphi(m)$ beyond the limit.
- The structural layer never asserts nonexistence; `NotFound` is returned only when the general descent was disabled.
- Root indices of non-real roots are exchanged between Wolfram and Python by value, not by index: Python orders conjugate pairs by real part and imaginary magnitude, and Wolfram's ordering of non-real roots can depend on its isolation method.

12 Conclusion

The restriction of `ToRadicals` is one of scope, not of principle. A radical expression exists exactly when the Galois group of the minimal polynomial is solvable, which is decidable in polynomial time; constructing the expression may require the splitting field, and this is where a general library function stops. With the splitting field available in exact coordinates, the Galois–Kummer descent is a short, fully proved, complete algorithm, and the structural reductions give the short formulas one wants in practice, including the two of the question. Both are implemented, tested and measured in two systems.

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